University of Glasgow

Title: DC-DC Boost Converter Operational Analysis

1. Objective:

To analyse the operation of boost converters using LTSpice simulation software and experiencing the actual boost converter. The objective is to understand the working principles, voltage regulation characteristics and inductor current of DC-DC boost converter. Students will compare the calculated values with simulation results.

Note: You must take the screen shots of every measurement you do through simulation and take a picture of outputs from multi meter or oscilloscope. The screen shots and pictures should be used in your lab report.

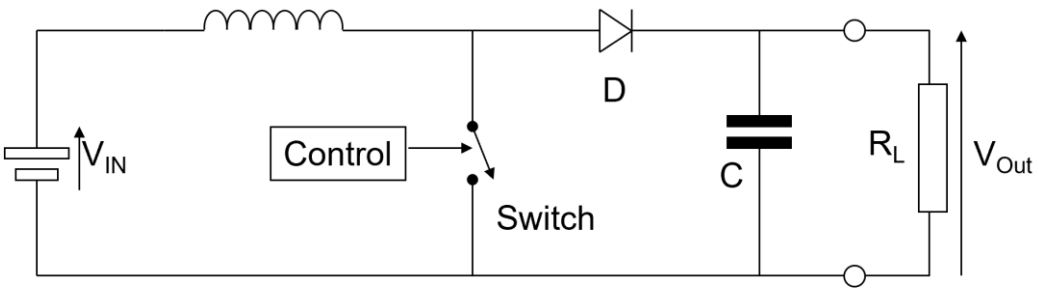
2. Introduction:

DC-DC converters are components in power electronics, enabling efficient voltage regulation and energy conversion. Boost converters is one of the fundamental topologies widely used in various applications.

Boost converter steps up the input voltage to a higher output voltage. It operates by controlling the duty cycle of a high-speed switch (typically a MOSFET) to regulate the output voltage efficiently. It uses the principle of inductor energy storage during the ON state and energy transfer to the output during the OFF state.

Circuit Diagram:

The circuit diagram of a buck converter is shown below:



Important characteristic parameters:

The voltage gain of the boost converter is given by:

$$V_{OUT} = V_{IN} \frac{1}{(1-D)} \quad \ldots\ldots (1)$$

Where:

- V_{OUT} = Output Voltage
- V_{IN} = Input Voltage
- D = Duty Cycle of the PWM signal (0 < D < 1)

Calculating the Duty Cycle (D)

The Duty Cycle for the boost converter is calculated using the equation:

$$D = 1 - \frac{V_{IN}}{V_{OUT}} \quad \ldots\ldots\ldots (2)$$

Assume a desired output voltage. For example:

- Input Voltage, V_{IN} = 12V

- Desired Output Voltage, V_{OUT} = 24V

Then,

D = 0.5

This means the switch will be ON for 50% of the time in each cycle.

Calculating the Output Voltage Ripple

The output voltage ripple value in the output of the boost converter is given by following expression:

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{DT}{RC} \ldots\ldots\ldots (3)$$

Calculating the Min/Max Inductor Current

The min and max inductor current is given by following expression:

$$I_{max} = I_L + \frac{\Delta I_L}{2} = \frac{V_{IN}}{(1-D)^2 R} + \frac{V_{IN} DT}{2L} \ldots\ldots\ldots (4)$$

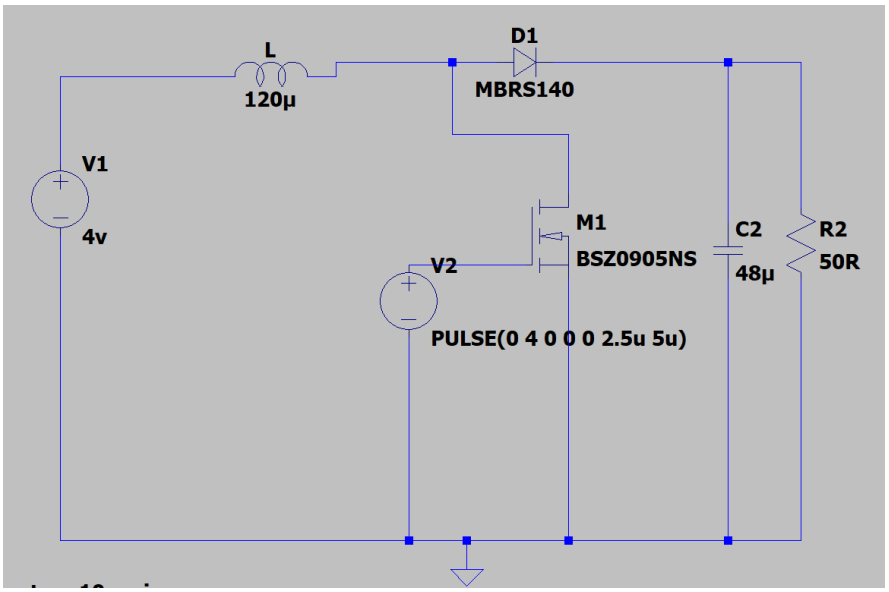
$$I_{min} = I_L + \frac{\Delta I_L}{2} = \frac{V_{IN}}{(1-D)^2 R} - \frac{V_{IN} DT}{2L} \ldots\ldots\ldots (5)$$

3. Experimental Work

Experiment 1: SPICE Simulation of a Buck Converter

Task-I

Implement the following circuit of a Boost Converter in SPICE using the components mentioned below.



- **Input Voltage (V1):** 4V
- **Inductor (L):** 120 µH
- **Diode (D1):** MBRS140
- **MOSFET (M2):** BSZ0905NS
- **Output Capacitor (C2):** 48 µF
- **Load Resistor (R2):** 50 Ω
- **PWM Pulse Source (V2):** PULSE(0 4 0 0 0 2.5µs 5µs)

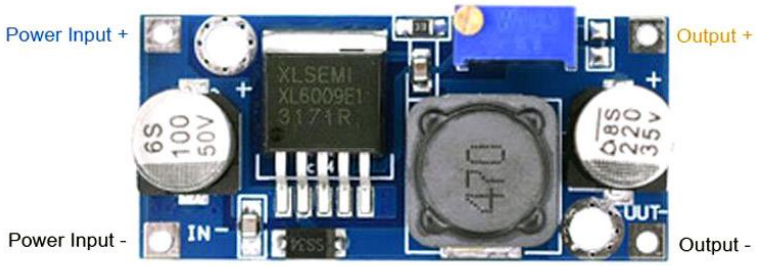
Task-II

Run the simulation and monitor the following outputs. Record the results and take the screen shots to put in the lab report.

1. Switching signal coming from V2 – Note time-period, ON time, duty cycle and frequency of V2.
2. The output voltage and compare it with the value calculated from expression given in equation 1.
3. The output voltage ripple value and compare it with the value calculated from equation 3.
4. Min and max value of inductor current and compare it with the value calculated from equation 4 and 5.
5. Now change the duty cycle to 25% and repeat step 1 to 4.

Task-III

In this step you will be using XL6009E1 Step-Up Adjustable DC-DC Switching Boost Converter.



1. Connect 5 Volts using power supply to the XL6009E1 input side.
2. Adjust the value of potentiometer to get a voltage of 10V. In this step output will be monitored using a multi meter.
3. Measure the ripple values in the output.
4. Now make the output voltage to 20V and measure it. To do so, repeat step 2 and 3.